

HAT-P-19b

R-1441

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_230915 · created 2026-08-01T08:26:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460202.7421 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.011
Transit depth (Rp/Rs)^2	1.89 +/- 0.29 [%]
Orbital Inclination (inc)	89.96 +/- 0.64
Airmass coefficient 1 (a1)	1.005 +/- 0.014
Airmass coefficient 2 (a2)	-0.0045 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	#3 - [244, 245]
Optimal Aperture	4.19
Optimal Annulus	10.39
Transit Duration (day)	0.1199 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.011 vs 0.1418 (deviation 0.4 σ)
Duration fit vs published	0.1199 d vs 0.1196 d (deviation 0.14 σ)
Mid-time O-C	12.0 min
Depth SNR	11.4
Residual scatter	1.44 %

Light curve

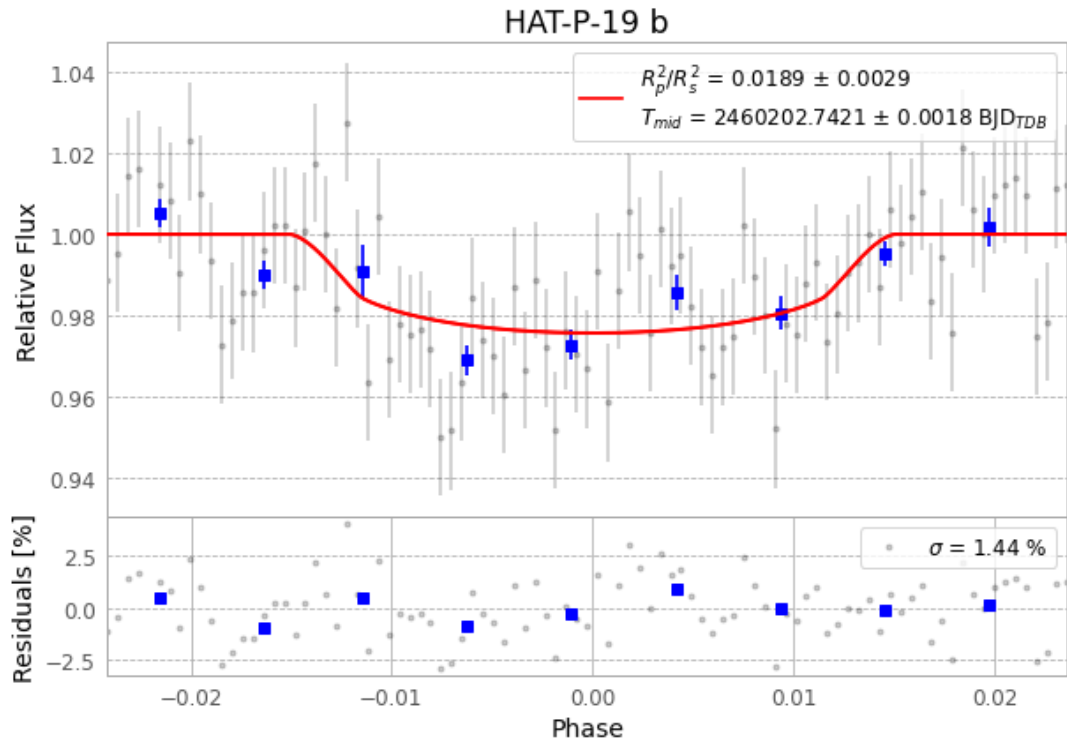


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-09-14.png (SHA-256 f675616ea08ae4f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d806b53f8ee0a1cf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

93 FITS frames (61 MB), HATP-19230915032115.FITS → HATP-19230915075715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-230915.log (5568d652797a...), FinalLightCurve_HAT-P-19 b_2023-09-14.png (f675616ea08a...), FinalLightCurve_HAT-P-19 b_2023-09-14.csv (d1257810a99c...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 08:26Z.